

Nonlinear biases in the roughness of a Fractional Stochastic Regularity Model

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ABSTRACT

A Multifractional Process with Random Exponent (MPRE) is used to model the dynamics of log-prices in a financial market. Under this assumption, we show that the Hurst–Hölder exponent of the MPRE follows the fractional Ornstein–Uhlenbeck process which in the Fractional Stochastic Volatility Model of Comte and Renault (1998) describes the dynamics of the log-volatility. We provide evidence that several biases of the estimation procedures can generate artificial rough volatility in surrogated as well as real financial data.

1. Introduction and motivation

In recent years, a debate is spreading in finance about the rough nature of the volatility process. The starting point is represented by the Fractional Stochastic Volatility Model (FSVM) introduced by Comte and Renault [1]

$$\begin{cases} dS_t = \mu(t, S_t)dt + S_t\sigma_t dW_t \\ d\ln(\sigma_t) = \theta[\mu - \ln(\sigma_t)]dt + \gamma dB_t^H, \sigma_0 > 0 \end{cases} \quad (1)$$

which models the logarithm of the volatility σ_t of stock price S_t as a fractional Ornstein–Uhlenbeck (fOU) process (with parameters θ , μ and γ) driven by a fractional Brownian motion (fBm) B_t^H independent on the Brownian motion W_t . When $H \neq 1/2$ fBm is not a semimartingale, thus neither the instantaneous volatility $\ln(\sigma_t)$ will be. However, the fact that this specification allows the volatility process to display long-memory does not affect model (1), which is indeed a semimartingale [1], consistently with a reasonable financial paradigm.

Using high-frequency data, Gatheral et al. [2] conclude that at any reasonable timescale the Hurst exponent H in (1) is very low (with values between 0.059 and 0.206), and propose the stationary Rough Fractional Stochastic Volatility (RFSV) model:

$$\begin{cases} dS_t = \mu_t S_t dt + S_t \sigma_t dW_t \\ \sigma_t = \exp(X_t) \end{cases} \quad (2)$$

where μ_t is a proper drift term and X_t is a fOU process satisfying

$$dX_t = \alpha(m - X_t)dt + \rho dB_t^H, \quad (3)$$

with $m \in \mathbb{R}$, ρ and α positive parameters and with W_t and B_t^H correlated in general. Several contributions (see, e.g. [3–7]) find evidence consistent with the hypothesis that time series of realized measures of

volatility are both rough and very persistent. In the same vein, [8] proves that non-rough volatility models are inconsistent to a power law of volatility skew and that, given a power law of volatility skew in an option market, a continuous price dynamics of the underlying asset with non-rough volatility admits an arbitrage. On the other hand, other recent contributions have questioned the above results and suggested that the apparent roughness observed in realized volatility can be ascribed to microstructure noise rather than the volatility process itself [9] or that simpler models than (2) can explain the patterns observed for volatility [10]. Moreover, it is precisely the measurement of volatility at high frequencies that is being questioned, because of both the noise emerging at these timescales and the fact that, generally, relevant financial problems arise at timescales larger than minutes (e.g., days, weeks, or months). It is primarily at these timescales that a model's ability to explain empirical data should be evaluated [10].

This paper contributes to the ongoing debate on volatility dynamics along two directions:

- (modeling) we replace the stochastic process in the first line of Eq. (1) by assuming that the log-price can be described by a Multifractional Process with Random Exponent (MPRE) [11–13]. Roughly speaking, this can be thought as a fractional Brownian motion (fBm) whose Hurst exponent randomly changes with respect to time t and path ω . If the change is continuous with respect to time, a local asymptotical property holds which ensures that in the neighborhood of any point t the MPRE behaves as a fBm. Since a linear relation can be stated between the logarithm of variance and the Hurst exponent of a fBm, we claim that the stochastic Hurst–Hölder parameter of a MPRE can replace the log-volatility in the second equation of model FSVM (1) [resp. (2)]. This sets up what can be called a *Fractional*

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Stochastic Regularity Model (FSRM). Our posit is strongly consistent with the empirical results that find stationarity, mean reversion to the value $1/2$, nearly normal distributions and presence of autocorrelation in the estimated sequences of the time-changing Hurst-Hölder exponent [14–18]. Actually, these very characteristics led us, since 2012 [19], to surrogate real financial time series by using a former version of MPRE [20], whose Hurst-Hölder function was modeled by a properly scaled fBm with a very small Hurst parameter. This is consistent with model (2), because the fBm and the fOU share very similar covariance functions, as pointed out by [21]. The model we propose is also in accordance with the empirical findings in [22], where the negative dependence found between prices' multiscaling and the Hurst exponent of the volatility process pushes the authors to foreshadow as a potential development the combination of multiscaling models for the price fluctuations and a fractional process for the volatility dynamics.

The benefit of writing the volatility process in terms of a time-changing Hurst-Hölder exponent modeled by a fOU process whose parameters are a transform of those in model (1) is to endow the notion of volatility with a deeper financial insight ensured by the relation linking the Hurst-Hölder exponent to the absence of arbitrage (this issue will be examined in paragraph 2.1).

- (estimation) we show that highly nonlinear biases, related to the time-lag, to the choice of some estimation parameters, to the estimator's own variance, and to the range of the fOU process, can affect the estimates of the global Hurst exponent of the stochastic Hurst-Hölder exponent (and, hence, of the log-volatility) and can lead to artificial roughness. Obviously, this does not mean that the regularity (volatility) process is not rough, but only that more robust estimation techniques should be used to conclude in favor of the roughness in an unambiguous way.

The remainder of the paper is organized as follows. In Section 2, after recalling some relevant properties of the fBm and the MPRE, the linear relationship between the instantaneous volatility and the pointwise Hurst-Hölder exponent of the MPRE is stated and its financial meaning is discussed. Section 3 deals with the estimation of the Hurst-Hölder local exponent H_t . In Section 4 the explicit representation of H_t by the fOU process is stated and a procedure is provided to assess the bias affecting the estimates of the parameter H of the fOU process. Section 5 exhibits the results of some simulations and show how biased the estimation of H can be. Finally, Section 6 concludes.

2. The model

Since early '80s, many controversial results about the Brownian nature of stock prices (see e.g. [23] for a review) led to consider models alternative to the Brownian motion, cornerstone of the whole asset pricing theory and reference process within the Efficient Market Hypothesis (EMH) paradigm.¹ Among the others, the fractional Brownian motion (fBm) with Hurst parameter $H \in (0, 1)$ [25] was introduced to account for positive or negative autocorrelation. In [26] it is defined as a Gaussian process $\{B_t^{H,C}, t \geq 0\}$, with Hurst parameter H and scale parameter C , with mean 0, stationary and self-similar increments such that

$$\mathbb{E} \left[\left(B_t^{H,C} - B_s^{H,C} \right)^2 \right] = C^2 |t - s|^{2H} \quad \forall s, t \in \mathbb{R}^+. \quad (4)$$

¹ According to Fama [24] “[...] a market in which prices always fully reflect available information is called *efficient*”. Depending on the information set, different forms of market efficiency can be given: the *weak form*, if current prices are assumed to reflect all past information; the *semi-strong form*, if also public information (e.g. announcements of annual earnings, stock splits...) is considered; and the *strong form*, if also private information available to some investors is accounted for.

The variance of fBm at time 1 or, given the stationarity of the increments, the variance of the unit time increment is

$$\mathbb{E} \left[\left(B_1^{H,C} \right)^2 \right] = \mathbb{E} \left[\left(B_1^{H,C} - B_0^{H,C} \right)^2 \right] = \mathbb{E} \left[\left(B_{t+1}^{H,C} - B_t^{H,C} \right)^2 \right] = C^2, \quad (5)$$

$B_t^{H,C}$ can be represented through the Wiener integral

$$B_t^{H,C} = \frac{C V_H^{1/2}}{\Gamma(H + 1/2)} \int_{\mathbb{R}} \left((t-s)_+^{H-1/2} - (-s)_+^{H-1/2} \right) dW_s \quad (6)$$

where $x_+ = \max(x, 0)$ and $V_H = \Gamma(2H + 1) \sin(\pi H)$; Γ denotes the Gamma function and the measure dW is Gaussian and independently scattered.

Since the process is self-similar, it follows that for the discretized sample X_j^H ($j \in [1, n]$) of fBm on the interval $t \in [0, 1]$, with $t = \frac{j-1}{n-1}$ one has that

$$\sigma^2(n) = \text{Var}(X_{j+1}^H - X_j^H) = C^2 n^{-2H} \quad (7)$$

that can be linearized as

$$\log \sigma(n) = \log C - H \log n. \quad (8)$$

fBm has:

- (i) Hölder continuous paths of order α , for any $\alpha \in (0, H)$,
- (ii) $\frac{1}{H}$ -variation on any time interval $[0, t]$ equal to $\mathbb{E}(|B_t^{H,C}|^{1/H}) t$ [27, 28],
- (iii) pointwise and uniform Hölder exponents almost surely equal to the Hurst exponent H [29,30].

Thus, the Hurst parameter dictates the regularity of the paths, in the sense that its trajectories are increasingly smoother for increasing values of H . From condition (ii) it follows that the fBm is not a semimartingale except for $H = 1/2$, when it reduces to a Brownian motion.

Since a constant Hurst parameter is too limiting for many real applications, fBm has been generalized to let H change through time and define new processes of the form $B_t^{H_t,C}$, with deterministic or stochastic H_t satisfying some Hölder regularity properties.² One of the generalizations are the *Multifractional Processes with Random Exponent* (MPRE) [12,13,20,38,39]. In general, they are continuous real-valued stochastic processes with non-stationary increments whose local path roughness can be prescribed via a functional Hurst parameter. More precisely, given a generic multifractional process $\{Z_t, t \geq 0\}$, its path roughness at any fixed $\tau > 0$ is usually measured through the pointwise Hölder exponent at τ , defined as (for details see [11,13]):

$$\alpha_Z(\tau) := \sup \{ \alpha \in [0, 1] : \limsup_{r \rightarrow 0^+} r^{-\alpha} \text{Osc}_Z(\tau, r) < +\infty \} \quad (9)$$

where, for all real number $r > 0$ small enough,

$$\text{Osc}_Z(\tau, r) := \sup \{ |Z_{t'} - Z_{t''}| : (t', t'') \in [\tau - r, \tau + r]^2 \}$$

is the oscillation of Z on the circular neighborhood of τ with radius r .

A general representation of the MPRE can be found in [12,13]. Given the filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in \mathbb{R}}, \mathbb{P})$, let $\{W_s, s \in \mathbb{R}\}$ denote the standard Brownian motion w.r.t. $\mathcal{F}_t$ and, for each $t \geq 0$, let $g_s(t)$ be $\mathcal{F}_s$ -adapted such that $g_s(t) = 0$ for $s > t$ with $\int_{-\infty}^t |g_s(t)|^2 ds < \infty$ for all t . The MPRE Z_t admits the following moving average representation

$$Z_t = \int_{-\infty}^t g_s(t) dW_s. \quad (10)$$

Function $g_s(t)$ is required to satisfy the following conditions (to ease comparison, the conditions are denoted using the same letters as in [12]):

² Such generalizations include, e.g., the multifractional Brownian motion (mBm) [29,31] and its Generalized version (GmBm) [32], the Multifractional Processes with Random Exponents (MPRE) [20], the bifractional Brownian motion [33], the mixed fBm [34], the fractional Riesz-Bessel motion [35], the Multi-fractional Generalized Cauchy Process [36] (see [37] for a review).

(A) Function $t \rightarrow g_s(t)$ is differentiable in $t > s$, for all s . There exist $\mathcal{F}_s$ -adapted processes H_s , L_s and R_s , such that $H_s \in (0, 1)$, $L_s > 0$, $R_s > 1/2$ and for all $t \geq 0$,

$$A1. |g_s(t)| \leq L_s |t - s|^{H_s - 1/2}, \quad \forall s \in (t - 1, t)$$

$$A2. |\partial_t g_s(t)| \leq L_s |t - s|^{H_s - 3/2}, \quad \forall s \in (t - 1, t)$$

$$A3. |\partial_t g_s(t)| \leq L_s |t - s|^{-R_s}, \quad \forall s \in (-\infty, t - 1].$$

(L-BC) There exists a sequence of stopping times $\tau_n \rightarrow \infty$, real numbers $\underline{H}_n$, $\overline{H}_n$, $\underline{L}_n$, $\underline{R}_n$, and a sequence of continuous, increasing functions $\omega_n(h)$ with $\omega_n(0) = 0$, such that

$$L\text{-BC1. } |H_{(t+h) \wedge \tau_n} - H_{t \wedge \tau_n}| \leq \omega_n(h),$$

$$L\text{-BC2. } 0 < \underline{H}_n \leq H_{t \wedge \tau_n} \leq \overline{H}_n < 1,$$

$$L\text{-BC3. } |L_{t \wedge \tau_n}| \leq \underline{L}_n \text{ and}$$

$$L\text{-BC4. } |R_{t \wedge \tau_n}| \geq \underline{R}_n > \frac{1}{2}.$$

Condition L-BC1 requires that H_t is continuous, what is satisfied by a continuous Itô process, a fBm or even a fOU. Condition L-BC2 is trivial under the assumption that H_t is a fOU. The local boundedness assumptions L-BC3 and L-BC4 are satisfied by either continuous processes or processes with bounded jumps.

Among the functions satisfying the above assumptions (A) and (L-BC) one can choose $g_s(t) := C[(t - s)_+^{H_s - 1/2} - (-s)_+^{H_s - 1/2}]$, so that a special case of the general MPRE process (10) is

$$B_t^{H_t, C} = C \int_{-\infty}^t \left[(t - s)_+^{H_s - 1/2} - (-s)_+^{H_s - 1/2} \right] dW_s. \quad (11)$$

This naturally leads to define the *Fractional Stochastic Regularity Process* (FSRP) as

$$\begin{cases} S_t = B_t^{H_t, C} \\ H_t = m' + \rho' \int_{-\infty}^t e^{-\alpha(t-s)} dB_s^H \end{cases} \quad (12)$$

where S_t denotes the log-price of a stock or an index, H_t is the unique pathwise solution of the fOU process (2) and parameters m' and ρ' are linear transforms of m and ρ of Eq. (2), as stated in Section 4, Eq. (35).

Processes $B_t^{H_t, C}$ and B_s^H are correlated through the parameter H_s . Assuming that the scale parameter of the fBm in the second line of Eq. (12) is 1 and exploiting Itô's isometry, one has

$$\begin{aligned} \text{Cov}(B_t^{H_t, C}, B_s^H) &= \mathbb{E} \left(B_t^{H_t, C} B_s^H \right) - \mathbb{E} \left(B_t^{H_t, C} \right) \mathbb{E} \left(B_s^H \right) = \mathbb{E} \left(B_t^{H_t, C} B_s^H \right) \\ &= C \mathbb{E} \left(\int_{-\infty}^{\infty} \left[(t - r)_+^{H_r - 1/2} - (-r)_+^{H_r - 1/2} \right] \left[(s - r)_+^{H_r - 1/2} - (-r)_+^{H_r - 1/2} \right] dr \right). \end{aligned}$$

When H_t is stationary, as in the case of the fOU process used in (12) to model it, the above covariance can be simplified as shown in Proposition 3.6 of [12]. Toilsome computations show that the above expected value can be written as

$$\begin{aligned} \mathbb{E} \left(B_t^{H_t} B_s^H \right) &= D(H_t, H) \left[|t|^{H+H_t} \cos \left(\pi \left(\frac{H - H_t}{2} - \text{sign}(t) \frac{H + H_t}{2} \right) \right) \right. \\ &\quad + |s|^{H+H_t} \cos \left(\pi \left(\frac{H - H_t}{2} - \text{sign}(s) \frac{H + H_t}{2} \right) \right) \\ &\quad \left. + |t - s|^{H+H_t} \cos \left(\pi \left(\frac{H - H_t}{2} - \text{sign}(t - s) \frac{H + H_t}{2} \right) \right) \right], \end{aligned}$$

$$\text{where } D(H_t, H) = \frac{\Gamma(H_t + 1/2) \Gamma(H + 1/2) \Gamma(2 - H - H_t)}{\pi(H + H_t)(1 - H - H_t)}.$$

To see why models (2) and (12) are related, one should consider that under a mild technical condition,³ at any time t_0 the pointwise

³ The condition is $\sup_{t \in [0, 1]} H_t < \beta_H([0, 1])$ where $\beta_H(J) = \sup \left\{ \beta : \sup_{s, s' \in J} \frac{|H_s - H_{s'}|}{|s - s'|^\beta} < \infty \right\}$ is the uniform Hölder exponent over the non degenerate (i.e. neither empty nor a singleton) compact interval J .

Hölder exponent of $B_{t_0}^{H_{t_0}, C}$ equals H_{t_0} almost surely, and $B_t^{H_t, C}$ fulfills the following locally asymptotic self-similarity property:

$$\lim_{\epsilon \rightarrow 0^+} \left(\frac{B_{t+\epsilon u}^{H_{t+\epsilon u}, C} - B_t^{H_t, C}}{\epsilon^{H_t}} \right)_{u \in \mathbb{R}^+} \stackrel{d}{=} \left(B_u^{H_t, C} \right)_{u \in \mathbb{R}^+}, \quad (13)$$

where $\stackrel{d}{=}$ denotes the equality among the finite dimensional distributions. Eq. (13) states that in the neighborhood of any point t , $B_t^{H_t, C}$ behaves like a fBm with Hurst exponent H_t . If also condition (A) is satisfied and assuming that the unit time variance of the fBm tangent to the MPRE is constant and equal to C^2 , at each point t

$$\log \sigma_t(n) = \log C - H_t \log n,$$

that is

$$H_t = p_1(n) \log \sigma_t + p_2(n), \quad (14)$$

with $p_1(n) = -\frac{1}{\log n}$ and $p_2(n) = \frac{\log C}{\log n}$. Eq. (14) states that H_t can be modeled by the same fOU which in model (2) provides the log-volatility, once a proper linear transforms is used for the parameters m and ρ of the model.

Furthermore, property (13) suggests that the estimation of the local H_t can be obtained by using a modification of variation statistics originally introduced to estimate the constant parameter H of a fBm (see Section 3).

2.1. Volatility estimation through the Hurst-Hölder exponent

The fact that conditions (i)–(iii) characterize the martingale property through the value of the Hurst-Hölder exponent provides the justification to express the volatility process in terms of the Hurst exponent itself. This has some advantages:

- unlike volatility, H_t is a measure of regularity and not of dispersion; therefore, the Hurst-Hölder exponent is sensitive to autocorrelation whatever the scale parameter, whereas volatility is not: using proper scale parameters, highly or not at all correlated data can have the same volatility, which seems unreasonable if volatility is intended as a measure of financial risk. In contrast, highly or not at all correlated data have different Hurst-Hölder parameters, because it is the intensity of autocorrelation that changes the smoothness of the process trajectories. This is particularly true for any zero-mean, stationary Gaussian processes X , which is α -Hölder continuous if its autocorrelation function ρ_X at zero is such that $1 - h^{2\alpha-\epsilon} \leq \rho_X(h) \leq 1 - h^{2\alpha+\epsilon}$ for $h \geq 0$ and for all $\epsilon > 0$ [40];
- volatility is a relative, not an absolute value. We can assess whether the market is more or less volatile today than it was yesterday, or a month ago, but we cannot argue what is the “fair” level of volatility consistent with a market “efficient” in the sense stated by the EMH. By contrast, the Hurst-Hölder parameter ranges in the interval (0,1) and equals 1/2 only if the process is a Brownian motion, which is the paradigm of the EMH. As described in [41], the distance $|H_t - 1/2| \in (0, 1/2)$ seizes the intensity of inefficiency that characterizes the market at each time t .
- when financial prices are modeled through processes that locally behave as a fBm (it is the case of MPRE under proper specification of H_t), volatility is related to H through Eq. (8), hence there is no loss of information in using the Hurst-Hölder parameter in place of volatility;
- in the last years many empirical estimates (see, e.g., [14–18]) have provided evidence that the time-varying Hurst-Hölder exponent of financial time series exhibits stationarity, mean-reversion, and nearly normal distribution around the value 0.5. All these patterns make the (fractional) Ornstein–Uhlenbeck process the natural candidate for modeling the dynamics of H_t . Therefore, it is quite natural to close the loop opened with relation (14), by assuming H_t precisely as the

fOU process appearing in the second equation of the model (1) or in Eq. (3).

Eq. (8) and the above arguments motivate us to study the roughness of volatility in terms of roughness of the Hurst-Hölder exponent. For this very reason, in the next section we will describe a class of estimators of H_t and then we will discuss how they affect the estimated roughness.

3. Estimation of the Hurst-Hölder exponent

The regularity of a stochastic process can be quantified following mainly two approaches: one can look at how it changes in time along the path (which is the case of the pointwise regularity estimated by the *local Hurst exponent*), or one can look at the overall scaling properties of the path, by calculating what is called the *generalized Hurst exponent* $H(q)$, which depends on the order q of the renormalized absolute moment of the lagged increments. Following the former approach, the process is called *multifractal* if the regularity changes through time; following the latter approach, the process is called *multifractal* if $H(q)$ is a non-linear function of q . Since the MPRE is both multifractal and multifractal, in the following both approaches will be considered to study the potential biases on the estimation of the level of roughness of the volatility.

3.1. Local Hurst exponent (LHE)

Variation statistics have been extensively used to estimate the local Hurst-Hölder exponent of time series.⁴ Detailed discussions, to which we refer, can be found in [26,42–44]. Given the discretized sample $(X_i, i \in \llbracket 1, n \rrbracket)$ of MPRE on the time domain $[0, 1]$, with time $t = \frac{i-1}{n-1}$, by virtue of limit (13), one has that within a window of sufficiently small length $\nu \ll n$ in which regularity is assumed to be constant

$$\frac{1}{\nu} \sum_{j=i-\frac{\nu-1}{2}}^{i+\frac{\nu-1}{2}} |X_{j+1} - X_j|^2 \sim \mathcal{N}\left(0, C^2 \left(\frac{1}{n-1}\right)^{2H_t}\right) \quad (15)$$

with $i = \left\lceil \left[1 + \frac{\nu-1}{2}, n - \frac{\nu-1}{2}\right] \right\rceil$. Since the k th absolute moment of a normal r.v. $Y \sim \mathcal{N}(0, \sigma^2)$ equals $\mathbb{E}(|Y|^k) = \frac{2^{k/2} \Gamma((k+1)/2)}{\Gamma(1/2)} \sigma^k$, the quantity

$$M_t^{(k)} = \frac{1}{\nu} \sum_{j=i-\frac{\nu-1}{2}}^{i+\frac{\nu-1}{2}} |X_{j+1} - X_j|^k, \quad k > 0 \quad (16)$$

satisfies

$$\mathbb{E}\left(M_t^{(k)}\right) = \frac{2^{k/2} \Gamma((k+1)/2)}{\Gamma(1/2)} C^2 \left(\frac{1}{n-1}\right)^{2H_t}. \quad (17)$$

In particular, when $k = 2$ one has

$$\mathbb{E}\left(M_t^{(2)}\right) = C^2 \left(\frac{1}{n-1}\right)^{2H_t}, \quad (18)$$

$$\text{with } \lim_{\nu \rightarrow \infty} P\left(\frac{M_t^{(2)}}{\mathbb{E}(M_t^{(2)})} = 1\right) = 1.$$

Since the scale parameter C is unknown, one way to get rid of it is first to calculate the quantity

$$M_t'^{(2)} = \frac{2}{\nu} \sum_{j=i-\frac{\nu-1}{2}}^{i+\frac{\nu-1}{2}} |X_{2j+1} - X_{2j-1}|^2 \quad (19)$$

and then observe that, as $n \rightarrow \infty$

$$\lim_{\nu \rightarrow \infty} P\left(\frac{M_t'^{(2)}}{M_t^{(2)}} = 2^{2H_t}\right) = 1, \quad (20)$$

from which one has the estimator

$$\hat{H}_t^{2,\nu,n} = \frac{1}{2} \log_2 \frac{M_t'^{(2)}}{M_t^{(2)}}. \quad (21)$$

Since estimator (21) is unbiased but with a low rate of convergence, [45] use it to correct the biased estimator of H_t introduced in [46] and refined in [19] which reads as

$$\hat{H}_t^{\nu,n,C^*} = -\frac{\log\left(\frac{1}{\nu} \sum_{j=i-\frac{\nu-1}{2}}^{i+\frac{\nu-1}{2}} |X_{j+1} - X_j|^2\right)}{2 \log(n-1)} + \frac{\log C^*}{\log(n-1)} \quad (22)$$

with C^* arbitrarily chosen.

More precisely, the following Proposition holds

Proposition 1. Let $\hat{H}_t^{2,\nu,n}$ and $\hat{H}_t^{\nu,n,C^*}$ as in (21) and (22), respectively. Then the estimator

$$\hat{H}_t^{(\nu)} = \hat{H}_t^{\nu,n,C^*} - h \quad (23)$$

with

$$h := \frac{\log(C^*/C)}{\log(n-1)} = \hat{H}_t^{\nu,n,C^*} - \hat{H}_t^{2,\nu,n} + \xi_t,$$

is unbiased, does not depend on C and has rate of convergence $\mathcal{O}(\nu^{-1/2}(\log n)^{-1})$.

Proof. Estimator (22) is biased because of the shift dictated by the difference $\frac{\log(C^*/C)}{\log(n-1)}$, C being the true parameter of sequence X .

Since estimator (21) is unbiased, denoted by ξ_t a random variable with mean zero, one can write

$$\hat{H}_t^{2,\nu,n} = H_t + \xi_t. \quad (24)$$

On the other side, it is also

$$\hat{H}_t^{\nu,n,C^*} = H_t + \frac{\log(C^*/C)}{\log(n-1)}. \quad (25)$$

Therefore,

$$h := \frac{\log(C^*/C)}{\log(n-1)} = \hat{H}_t^{\nu,n,C^*} - \hat{H}_t^{2,\nu,n} + \xi_t,$$

from which, by averaging with respect to the number of estimates, it follows

$$h = \frac{1}{n-\nu} \sum_{i=1+\frac{\nu-1}{2}}^{n-\frac{\nu-1}{2}} \left(\hat{H}_t^{\nu,n,C^*} - \hat{H}_t^{2,\nu,n} \right).$$

Since C^* is chosen arbitrarily, the corrected estimate is

$$\hat{H}_t^{(\nu)} = \hat{H}_t^{\nu,n,C^*} - h. \quad (26)$$

Since the correction turns to be a shift, the rate of convergence is the same as $\hat{H}_t^{\nu,n,C^*}$, which is equal to $\mathcal{O}(\nu^{-1/2}(\log n)^{-1})$ as proved in [46]. $\square$

When $H = 0.5$, $\hat{H}_t^{\nu,n,C}$ is normally distributed with mean 0.5 and variance equal to $\frac{1}{2\nu \log^2(n-1)}$ ([46], eq.(3.13)). Since the result continues to hold for $\hat{H}_t^{(\nu)}$, this allows to test the significance of the departures of H_t from the equilibrium value 0.5.

Notice that, because of (15), the quantity $\frac{1}{\nu} \sum_j (X_{j+1} - X_j)^2$ appearing in the numerator of the first fraction of Eq. (22) is an estimate of the variance σ_t^2 of the process increments at time t , for a window of size ν . Thus, denoting it by $\hat{\sigma}_{t,\nu}^2$, one can write

$$\hat{H}_t^{\nu,n,C^*} = -\frac{\log \hat{\sigma}_{t,\nu}^2}{2 \log(n-1)} + \frac{\log C^*}{\log(n-1)}$$

⁴ Along with the *pointwise* Hölder exponent defined by relation (9), the local Hölder regularity of the trajectory $t \mapsto Z_t$ in a neighborhood of some fixed point τ can be measured through the *local* Hölder exponent, defined as $\tilde{\alpha}_Z(\tau) = \sup\{\beta_Z([u, v]) : u, v \in \mathbb{R} \text{ and } \tau \in (u, v)\}$.

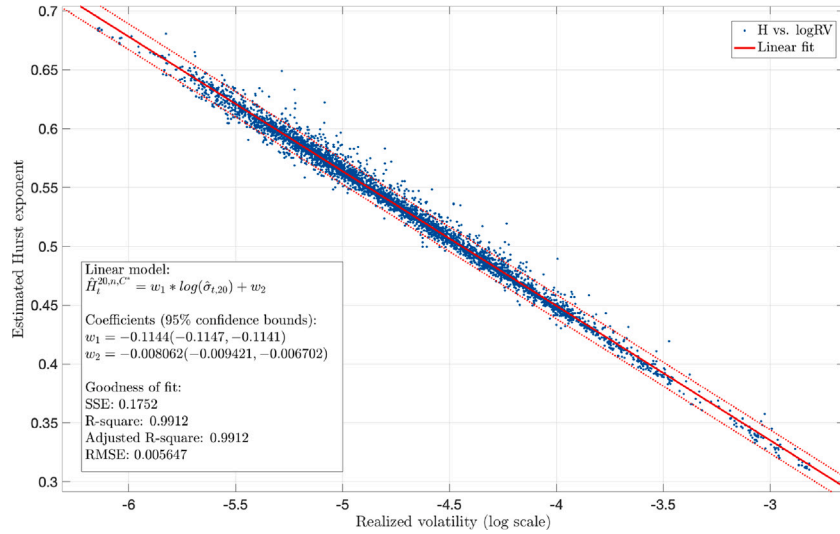


Fig. 1. Fit of relation (27) for the S&P500. The dotted lines display the 95% confidence bounds.

that is, setting $w_1 = -\frac{1}{\log(n-1)}$ and $w_2 = \frac{\log C^*}{\log(n-1)}$,

$$\hat{H}_t^{v,n,C^*} = w_1 \log \hat{\sigma}_{t,v} + w_2. \quad (27)$$

or, equivalently,

$$\hat{H}_t^{(v)} = w_1 \log \hat{\sigma}_{t,v} + w_3. \quad (28)$$

with $w_3 = w_2 - h$.

Eq. (27) [resp. (28)] restates for estimator (25) [resp. (26)] the general relationship (8) between volatility and regularity. As an example, Fig. 1 displays the linear fit (27) for the daily S&P500 closing data from May, 3rd 1978 to December, 31st 2017 (10,005 observations). The fit returns an R-square equal to 0.9912, as expected by the analytical nature of the relation.

Remark 1. Estimator (26) is built for a time changing Hurst exponent. Nevertheless, when H is a constant (as in the case of fBm or of fractional Ornstein–Uhlenbeck process used in the Rough Fractional Stochastic Volatility model, see Section 4), a number of well-known global estimators are available, both in time and in frequency domains. Since in Section 4 it will be argued that the sequence $\{(t, \hat{H}_t^{(v)})\}$ behaves as a fractional Ornstein–Uhlenbeck process and it is therefore characterized by a constant H , in Section 5 its global Hurst exponent will be estimated using three different estimators: the *Absolute Moment method* (AM) [47], the *Aggregated Variance* (AV) [48] and *Higuchi's method* (HH) [49]. Since these are three well-known and widely used estimators, we refer to the references for details.

Remark 2. Unless vertical shifts due to the unknown scaling parameter C^* appearing in w_3 , Eq. (28) suggests that the daily Hurst–Hölder exponent can be estimated by the realized volatility. In fact, since this is calculated using intraday data, the length v of the estimation window becomes 0 if the realized volatility replaces the v -days volatility estimate $\hat{\sigma}_{t,v}$ in Eq. (28). The good news is that, being just a vertical shift parameter, w_3 does not affect the overall regularity of the sequence $\{(t, \hat{H}_t^{(0)})\}$. In other words, the fact that $\hat{H}_{\hat{H}_t^{(0)}}$ does not depend on w_3 allows to analyze the roughness of the (eventually shifted) Hurst–Hölder sequences even when they are estimated through the realized volatility. This analysis will be performed in Section 5.2 for comparison purposes with other works focused on realized volatility.

3.2. Generalized Hurst exponent (GHE)

Along with the estimator of the time-changing Hurst exponent, the multiscaling properties of the MPRE can be assessed by means of the Generalized Hurst Exponent, which is deduced by the scaling of the q th-order moments of the distribution of the increments. A complete description of the estimation method, introduced by [50], can be found in [51] or [52]. We will compute the generalized Hurst exponent $H(q)$ with the scaling $\zeta(q) = qH(q)$ of the renormalized q th absolute moment of the distribution of stationary increments of process X_t

$$\tilde{M}_{q,\tau} = \frac{\mathbb{E}[|X_{t+\tau} - X_t|^q]}{\mathbb{E}[|X_t|^q]} = K(q)\tau^{\zeta(q)}, \quad (29)$$

where τ is the temporal lag over which the increments are computed. For a given q and $\tau \in [\tau_{\min}, \tau_{\max}]$, $H(q)$ is derived from the slope of the linearization of (29)

$$\ln \left(\frac{\mathbb{E}[|X_{t+\tau} - X_t|^q]}{\mathbb{E}[|X_t|^q]} \right) = qH(q) \ln \tau + \ln(K(q)). \quad (30)$$

The parameter q is generally chosen to be less than the tail exponent α of the distribution of returns. Since financial time series are characterized by a tail exponent varying between about 1.5 and 3, [52–56] use a conservative approach considering $q \leq 1$. Furthermore, since the returns can be 0 and the moments of negative orders are not defined [53], we will also set $q > 0$. For the multiscaling analysis, all the estimates have been obtained setting $q = 1$, since this value is closely related to the Hurst exponent (see [22], Section 2).

The parameter τ can be chosen using several approaches [53–56]; among them, the *Segmented Regression on the structure function* [54] and the *Autocorrelation Significance test* [53] deserve to be mentioned. In the application of Section 5, for each simulation we will estimate $\tau_{\max}$ following the *Autocorrelation Segmented Regression method* (ACSR) recently discussed in Brandi and Di Matteo [56]. The method splits the autocorrelation function of the returns r_t in the two states

$$\rho_\tau(r_t) = \begin{cases} \alpha + \beta\tau, & \text{if } \tau < \tau_{\max} \quad (\text{High autocorrelation state}) \\ \alpha + \beta\tau_{\max}, & \text{if } \tau \geq \tau_{\max} \quad (\text{Random noise state}) \end{cases}$$

where α is the intercept and β is the slope of the regression. Taking $\tau_{\max}$ as the value which minimizes the sum of squared residuals for the high autocorrelation state and the random noise state, one maximizes the autocorrelation information.

Remark 3 (RNSGHE). Brandi and Di Matteo [56] show that the ratio

$$\tilde{\Xi}_{q,\tau} := \frac{\tilde{M}_{q,\tau}}{K(q)} = \tau^{\zeta(q)}, \quad (31)$$

eliminates the possible bias introduced by estimation of $K(q)$ via regression. By defining the q th-order normalized moment as

$$\tilde{\Xi}_{q,\tau} := (\tilde{\Xi}_{q,\tau})^{1/q},$$

Eq. (31) becomes

$$\tilde{\Xi}_{q,\tau} \sim \tau^{H(q)}. \quad (32)$$

With these transformations, the regressions have 0 intercept for all q 's and multiscaling is detected only if the regression coefficients $H(q)$ differ for distinct values of q . This approach, called *Relative Normalized and Standardized Generalized Hurst Exponent (RNSGHE)*, will be used in the application to compare the results obtained by the classical GHE method.

4. (Rough) fractional stochastic volatility

Studying the smoothness of the volatility process at high frequencies, [2] argues that the logarithm of realized volatility is rough and behaves as a fBm with Hurst exponent $H \in (0, \frac{1}{2})$. They consider the *Rough Fractional Stochastic Volatility* (RFSV) model (2) where X_t , $t \in [0, T]$, follows the fOU process (3), whose unique pathwise solution is

$$X_t = e^{-\alpha t} X_0 + (1 - e^{-\alpha t})m + \rho \int_0^t e^{-\alpha(t-s)} dB_s^H, \quad (33)$$

where X_0 is the initial value and the integral is a pathwise Riemann–Stieltjes integral (this can be seen combining partial integration with the finite variation of the exponential function). Cheridito et al. [21] prove that the integral in (33) is almost surely continuous in t and that the covariance function of X_t decays very similarly to that of the increments of B_t^H , which are recovered from (33) by setting $\alpha = 0$. When $\alpha < 0$, X_t is obviously explosive and $X_t = \mathcal{O}_p(e^{-\alpha t})$ as $t \rightarrow \infty$ [57].⁵ Finally, when $\alpha > 0$, Hu and Nualart [58] prove stationarity and ergodicity for X_t under the condition

$$X_0 = m + \rho \int_{-\infty}^0 e^{\alpha s} d\tilde{B}_s^H,$$

with $\tilde{B}_s^H = {}^1 B_s^H \mathbb{1}_{s \geq 0} + {}^2 B_{|s|}^H \mathbb{1}_{s < 0}$ a two-sided fBm with ${}^2 B_{|s|}^H$ independent of ${}^1 B_{|s|}^H$. Under this condition, X_t is a covariance stationary process with

$$\mathbb{E}(X_t) = m \quad \text{and} \quad \text{Var}(X_t) = \rho^2 \alpha^{-2H} H \Gamma(2H).$$

Moreover, X_t can be represented as

$$X_t = m + \rho \int_{-\infty}^t e^{-\alpha(t-s)} dB_s^H, \quad (34)$$

from which one can readily see that, when α is small enough with respect to T , the log-volatility behaves locally as a fBm. We expect the same behavior also when σ_t is replaced by H_t . In fact, combining the pathwise solution (34) with $X_t = \log \sigma_t$, it trivially follows that

Proposition 2. *Given limit (13) and the linearity of relation (14), the assumption that the stochastic volatility in models ((1) or (2)) follows a fOU process entails that the pointwise Hurst–Hölder exponent H_t of the MPRE also follows a fOU process with rescaled parameters, that is*

$$X_t = \log \sigma_t \stackrel{\text{by (14)}}{=} \frac{H_t - p_2}{p_1} = m + \rho \int_{-\infty}^t e^{-\alpha(t-s)} dB_s^H$$

⁵ The notation $X_t = \mathcal{O}_p(a_t)$ as $t \rightarrow \infty$ means that the set of values X_t/a_t is stochastically bounded, i.e. for any $\epsilon > 0$, there exists a finite $M > 0$ and a finite $\tau > 0$ such that $\mathbf{P}(|X_t/a_t| > M) < \epsilon$, $\forall t > \tau$.

$$H_t = m' + \rho' \int_{-\infty}^t e^{-\alpha(t-s)} dB_s^H \quad (35)$$

with $m' = p_1 \cdot m + p_2$ and $\rho' = p_1 \cdot \rho$.

Proof. Omitted, since it trivially follows from relation (14) implemented in the second line of model (1) or (2). $\square$

Of course, a similar argument holds approximately for the estimates $\log \hat{\sigma}_{t,v}$ and $\hat{H}_t^{v,n,C^*}$ (or $\hat{H}_t^{(v)}$). In this case one can readily check from (27) that

$$\begin{aligned} \hat{X}_t = \log \hat{\sigma}_{t,v} &= \frac{\hat{H}_t^{v,n,C^*} - w_2}{w_1} \approx m + \rho \int_{-\infty}^t e^{-\alpha(t-s)} dB_s^H \\ \hat{H}_t^{v,n,C^*} &\approx m'' + \rho'' \int_{-\infty}^t e^{-\alpha(t-s)} dB_s^H \end{aligned}$$

or, equivalently

$$\hat{H}_t^{(v)} \approx m''' + \rho'' \int_{-\infty}^t e^{-\alpha(t-s)} dB_s^H, \quad (36)$$

with $m'' = w_1 \cdot m + w_2$, $m''' = w_1 \cdot m + w_2 - h$ and $\rho'' = w_1 \cdot \rho$. Relation (36) states that the estimated H_t is expected to behave as a fractional Ornstein–Uhlenbeck process. This is in line with the idea suggested by Garcin [59], who – moving from the approach suggested by [15] – builds a model of nested fractality and estimates the Hurst exponent of the Hurst exponents for several FX rates between 2006 and 2016.

Relative to the RFSV model, $\log \sigma_t$ is modeled by a fOU driven by a given H . Many recent contributions estimate from financial data low values of H for the fOU process, which is precisely why stochastic volatility is given the adjective *rough*. Our aim is to investigate the discrepancy that the estimation procedures can induce between the prescribed (true) H and the estimated Global Hurst exponent $\hat{H}$, obtained by either the LHE or the GHE.

As for the LHE, the global H is the result of a twofold estimation procedure: the first provides the local exponent $\hat{H}_t^{(v)}$, while the second provides the global exponent $\hat{H}$ of the entire path $\{(t, \hat{H}_t^{(v)})\}$. Each local estimate is affected by the estimator's variance and each estimate of the global H depends on the size of the (non overlapping) blocks used to partition the sequence.

As for the GHE, the choice of the range of values for q as well as the value of τ (see Eq. (29)) can result in spurious changes of the slope in Eq. (30). In both cases, the combination of several factors introduces nonlinear biases in the estimates, resulting in a global $\hat{H}$ that can be artificially smaller or higher than the prescribed H . To show the bias that can affect the estimate of a global H , in the next section we will implement the following steps:

1. Given model (36), generate $H_t \in [H_{\min}, H_{\max}] \subset [0, 1]$ as a fOU process with a prescribed value $H \in (0, 1)$, and $m > 0$, $\alpha > 0$, $\rho > 0$, $\sigma > 0$.
2. Simulate an MPRE driven by H_t obtained in step 1.
3. Set v and estimate the LHE $\hat{H}_t^{(v)}$ from the MPRE.
4. Estimate the global $\hat{H}_{\hat{H}^{(v)}}$ from the sequence $\{(t, \hat{H}_t^{(v)})\}$, using the AM, AV and HH estimators.
5. Use trajectories built in step 3 to estimate the GHE.

5. Data analysis

In this section, we present some numerical simulations based on the steps 1–5 in Section 4. We will compare the Hurst exponent H prescribed in step 1 to simulate the fOU process with the global Hurst exponent, estimated following the procedures described in steps 4 and 5. The running parameters to simulate trajectories of length $n = 20480$ of the fOU process have been set equal to $\alpha = 1$, $m' = 0.5$, $\rho' = 0.5$, $\sigma = 1$ (see Eq. (35)).

The estimation of the global H from the LHE (step 4) is obtained applying the three estimators AM, AV and HH to the sequences $\{(t, \hat{H}_t^{(v)}), v = 8, 10, 15, 20, 30, 40, 50, 60, 80\}$. These methods

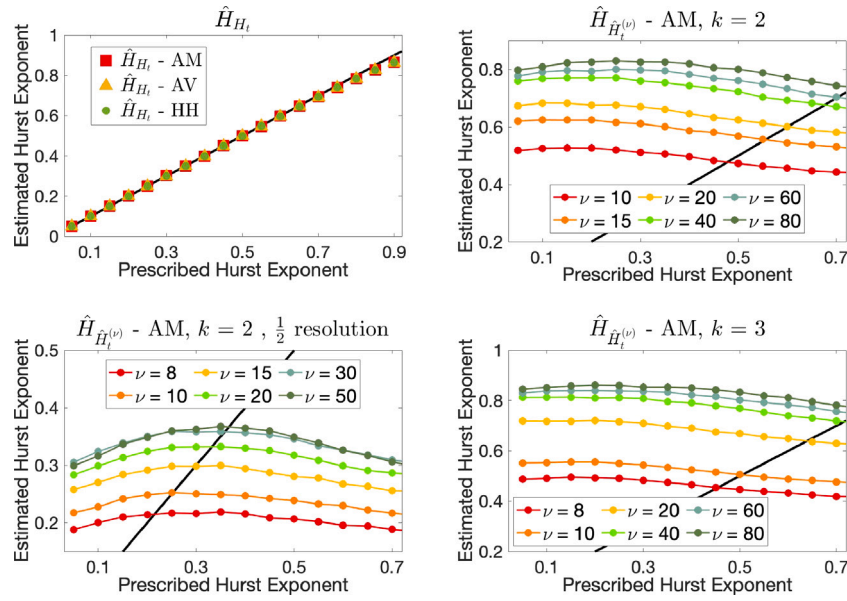


Fig. 2. (Top left panel) The estimates for Global Hurst exponent with AM, AV and HH methods, have a linear behavior following the black line, which indicates the bisector. Top right panel displays the global $\hat{H}_{\hat{H}_t^{(\nu)}}$ when $k = 2$. The bottom left panel displays $\hat{H}_{\hat{H}_t^{(\nu)}}$ when in Eq. (22) we halve the resolution of data. Finally, the bottom right panel displays the global $\hat{H}_{\hat{H}_t^{(\nu)}}$ when $k = 3$. In all cases, data are averaged over 100 simulations for different length ν .

divide the series into blocks of size m and average within each block (see [47]). We used three block sets: $m_1 = \{1, 2, 4, 8, 12, 16, 25, 32\}$, $m_2 = \{50, 100, 150, 200, 250, 300, 350\}$, $m_3 = \{5, 10, 20, 30, 40, 50, 60, 70\}$, where the maximal length of each block is significantly lower than the length of the trajectories. In addition, the comparison is performed by modifying some parameters as the co-domain $[H_{min}, H_{max}]$ of the fOU process H_t , the length ν and the order k of the LHE estimator. The GHE is estimated for each sequence $\{(t, \hat{H}_t^{(\nu)})\}$, $\nu = 10, 20, 40, 50, 60, 80$, with $q = \{0.01, \dots, 1; \text{step } 0.01\}$ and $\tau \in [1, \tau_{max}]$.

For reasons of space and to avoid redundancy (by changing the above parameters, similar results are achieved), only some of the plots are reproduced.

5.1. Global Hurst exponent from LHE

In the top left panel of Fig. 2, for all global estimators considered, the global estimated $\hat{H}_{H_t}$ for fractional Ornstein–Uhlenbeck agree with the prescribed H : in all cases we get a linear behavior. Then we studied the global $\hat{H}_{\hat{H}_t^{(\nu)}}$ for different amplitudes ν , using the local estimator described in Eq. (26) and varying some parameters. In particular, in the top right panel and the bottom right panel, we reported, with the variation of ν , the estimate of the global $\hat{H}_{\hat{H}_t^{(\nu)}}$, respectively for an order of variation $k = 2$ and $k = 3$. In the bottom left panel of Fig. 2, the same simulations are reported, but using halving the resolution in Eq. (22) to test for the stability of the results. In all these cases, we observe that the estimate of the global $\hat{H}_{\hat{H}_t^{(\nu)}}$ strongly depends on the parameter ν . The local estimator has a variance decreasing with the length of the window: the larger ν , the smoother the estimates $\hat{H}_t^{(\nu)}$, and therefore the larger the global Hurst exponent. We find non-linearity, non-injectivity and distance of the estimated global Hurst exponents with respect to the initial prescribed values of H ; in fact, absent biases, one should expect the estimates to lie on the black solid line, bisector of the plots, what is absolutely not.

Just as the local estimator depends on ν , the global Hurst estimators strongly depend on the chosen set of blocks. The top panel of Fig. 3 displays the nonlinear behavior of the global $\hat{H}_{\hat{H}_t^{(\nu)}}$ for AV, HH and AM methods with three different block sets: m_1, m_2, m_3 . Given a single block m_i , we observe that the curves resulting from different methods have a nonlinear but similar behavior, net to vertical shifts. For different

sets of blocks we note that the shape of the estimated curve varies significantly. Bottom panels in Fig. 3 relate the global estimated Hurst exponent to the range of the co-domain of the starting H_t (fOU centered ad 0.5). We observe that by simulating H_t in increasingly smaller intervals $[H_{min}, H_{max}]$, the global $\hat{H}_{\hat{H}_t^{(\nu)}}$ tends to decrease. Therefore, we see an artificial creation of roughness as the interval $[H_{min}, H_{max}]$ tightens. This effect is attributable to the variance of the local estimator, which predominates the more the variations of the H_t become small.

5.2. Realized volatility analysis

The previous Section documents that several factors (the size ν of the estimation window, the size of the blocks m_i , the range of the simulated H_t) bias the estimation of the overall regularity. In particular, the fact that spurious autocorrelations in the estimated data increase with ν poses a problem of arbitrariness in the degree of smoothness/roughness calculated by using volatility estimators based on moving windows. For this reason and to ease comparisons with works focused on realized volatility (i.e. the spot daily volatility measured using intraday data and not ν -days moving windows), we examine the regularity of the sequence $\{(t, \hat{H}_t^{(0)})\}$, where $\hat{H}_t^{(0)}$ is calculated using the realized volatility in Eq. (28) (see Remark 2). Then, regarded $\hat{H}_t^{(0)}$ as a path of the fOU process (35),⁶ we applied steps 2–4 outlined at the end of Section 4.

The realized volatility was calculated using the 5 minutes realized variance published by the Oxford-Man Institute's Realized Library, between January 3, 2000 and June 27, 2018 for the following stock indexes: Standard & Poor 500 (SPX), Dow Jones Industrial Average (DJI), Footsie 100 (FTSE), Hang Seng (HSI), Nikkei 225 (N225), Nifty 50 (NSEI) and Swiss Market Index (SSMI). Table 1 summarizes the global Hurst exponents estimated applying the AM, AV and HH methods (gray

⁶ While $\hat{H}_t^{(0)}$ does not depend on w_3 , the shift becomes essential when $\hat{H}_t^{(0)}$ is used to simulate paths of the MPRE. Therefore, to surrogate MPRE data, we forced the sequence $\hat{H}_t^{(0)}$ to have mean value equal to 0.5. Two reasons support this choice: first, the resulting MPRE is financially meaningful, in the sense that a mean of $\hat{H}_t^{(0)}$ different from 0.5 would entail systematic arbitrage opportunities; second, there is a nearly unanimous consensus of the empirical analyses that the long-term mean of $\hat{H}_t^{(0)}$ is very close to 0.5, whatever the financial time series.

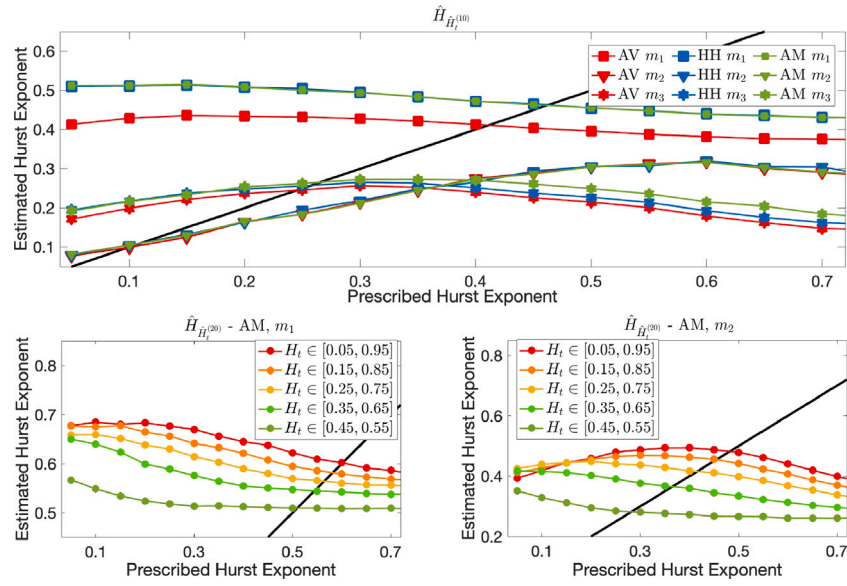


Fig. 3. Top panel: $\hat{H}_{\hat{H}_t^{(v)}}$ for AV, HH and AM methods with the three block sets m_1, m_2, m_3 . For each global estimator (AM, AV, HH), the estimates are nonlinear in the prescribed Hurst exponent. The shape of the curve changes according to the set of blocks used. Bottom panels: values of $\hat{H}_{\hat{H}_t^{(v)}}$ as the range of H_t (FOU centered at 0.5) varies for the block sets m_1 and m_2 . Each value is an average on 100 simulations.

Table 1

Global Hurst exponent $\hat{H}_{\hat{H}_t^{(v)}}$ for seven stock indexes (gray columns), compared with $\hat{H}_{\hat{H}_t^{(v)}}$ ($v = 5, 10, 20$), obtained by applying estimator (26) to the MPRE surrogated with random parameters given by the sequences $\{(t, \hat{H}_t^{(0)})\}$ centered on 0.5. Numbers in parenthesis represent the standard errors related to the estimated coefficients.

		$\hat{H}_{\hat{H}_t^{(v)}}$	$\hat{H}_{\hat{H}_t^{(5)}}$	$\hat{H}_{\hat{H}_t^{(10)}}$	$\hat{H}_{\hat{H}_t^{(20)}}$
AM	SPX	0.084	0.057(0.007)	0.094(0.006)	0.173(0.007)
	DJI	0.097	0.045(0.006)	0.109(0.006)	0.171(0.007)
	FTSE	0.127	0.071(0.007)	0.098(0.006)	0.173(0.007)
	HSI	0.056	0.075(0.006)	0.074(0.006)	0.168(0.007)
	N225	0.046	0.084(0.005)	0.107(0.005)	0.196(0.006)
	NSEI	0.136	0.061(0.006)	0.083(0.006)	0.144(0.007)
	SSMI	0.049	0.020(0.007)	0.054(0.007)	0.146(0.006)
AV	SPX	0.071	0.042(0.007)	0.073(0.006)	0.142(0.006)
	DJI	0.048	0.022(0.006)	0.074(0.006)	0.135(0.007)
	FTSE	0.087	0.028(0.006)	0.086(0.006)	0.136(0.006)
	HSI	0.059	0.052(0.006)	0.049(0.006)	0.126(0.007)
	N225	0.054	0.053(0.005)	0.103(0.005)	0.168(0.006)
	NSEI	0.137	0.056(0.006)	0.060(0.006)	0.123(0.007)
	SSMI	0.024	0.011(0.005)	0.033(0.007)	0.109(0.006)
HH	SPX	0.102	0.081(0.002)	0.120(0.003)	0.193(0.003)
	DJI	0.099	0.081(0.002)	0.114(0.002)	0.188(0.003)
	FTSE	0.112	0.037(0.002)	0.122(0.002)	0.156(0.002)
	HSI	0.103	0.077(0.002)	0.104(0.003)	0.190(0.003)
	N225	0.076	0.070(0.002)	0.096(0.002)	0.173(0.002)
	NSEI	0.076	0.068(0.002)	0.098(0.002)	0.164(0.002)
	SSMI	0.155	0.080(0.002)	0.118(0.002)	0.200(0.003)

columns) to the sequence $\{(t, \hat{H}_t^{(0)})\}$ of each index; in all cases we obtain low values of regularity. Centering the sequences on 0.5, we simulated paths of MPRE and estimated $\hat{H}_{\hat{H}_t^{(v)}}$, with $v = 5, 10, 20$.

As the window size decreases, the roughness increases for all indexes but HSI in the case of AM and AV; for these, anyway, the anomaly can be explained by a standard error which is larger than the positive difference as v decreases. In all the other cases the roughness reaches smaller values than those obtained from the realized volatility; this result, at first glance counterintuitive, is actually a consequence of the variance of estimator (26), which increases as v decreases, thus generating artificial roughness.

5.3. Global Hurst exponent from GHE

Multiscaling analysis is justified by the multifractality exhibited by the MPRE.

This is shown by Fig. 4: the scale parameter $\zeta(q) = q\hat{H}_{H_t}(q)$ is linear in q for the FOU process H_t , which is indeed unifractal with $\hat{H}_{H_t}(q) = H$ (blue dots). The linear fit returns in this case an R-square equal to 0.9999. By contrast, $\zeta(q) = q\hat{H}_{B_t^{H_t,C}}(q)$ is non linear in q when the input is the MPRE $B_t^{H_t,C}$ surrogated with random exponent equal to the simulated FOU process H_t (red triangles). Precisely the nonlinearity of $\zeta(q)$ indicates the multifractality of the MPRE.

Following steps 1–5 in Section 4, multiscaling analysis is used to compare $\hat{H}_{H_t}(1)$ and $\hat{H}_{\hat{H}_t^{(v)}}(1)$ as obtained from Eq. (30). In Fig. 5, data from nine multiscaling analyses are shown for different prescribed exponents H and lengths v , with $\tau \in [1, \tau_{\max}]$. In all cases, $\hat{H}_{\hat{H}_t^{(v)}}$ displays two distinct linear regions: one for $\tau \leq v$, the other for $\tau > v$. The window of length v of the local estimator described by Eq. (26) increases the autocorrelation between the data up to $\tau = v$. In the second region, autocorrelations are purged of the moving window bias and this reflects in a slope which is lower than the one observed for the region $\tau \leq v$.

To show the inconsistency between the prescribed H and the generalized $H(q)$ we estimated $\hat{H}_{\hat{H}_t^{(v)}}$ both for $\tau \in [1, \tau_{\max}]$ and for $\tau \in [v + 1, \tau_{\max}]$ and compared the two estimates with $\hat{H}_{H_t}$, averaging all data over 100 simulations. Fig. 6 displays the results: $\hat{H}_{H_t}$ (gray dotted line) has a linear behavior and follows the black line, which indicates the bisector (perfect match between prescribed and estimated H). Conversely, both the filtered and unfiltered estimates for $\hat{H}_{\hat{H}_t^{(v)}}$ follow a nonlinear trend, as in the case of the Global Hurst exponent from LHE. As already noted, $\hat{H}_{\hat{H}_t^{(v)}}$ with $\tau \in [1, \tau_{\max}]$ (blue triangles) is always smoother because of the spurious autocorrelations triggered by the moving window estimation. Even if it is almost trivial, we also observe that as the length v increases, the overall smoothness of the estimate increases. The results do not substantially change when the analysis is performed using the RNSGHE method, for which the patterns are virtually undistinguishable from the previous case (see Fig. 7).

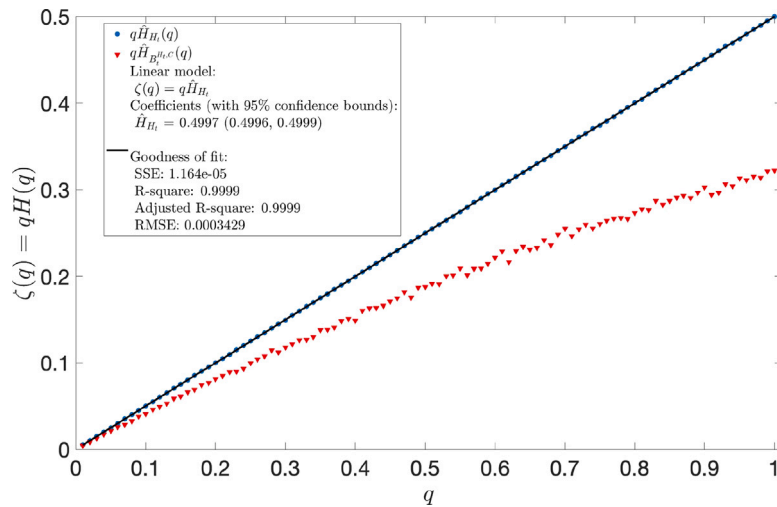


Fig. 4. The scale parameter $\zeta(q) = qH(q)$ is linear for the fOU, which is unifractal with $\hat{H}_H(q) = H$ (blue dots). The black line displays the linear fit with an R-square equal to 0.9999. On the other hand, a MPRE has multifractal properties. In fact, the generalized $\hat{H}_{B^{H,c}}(q)$ is dependent on the parameter q and $\zeta(q)$ is a non linear function (red triangles). Each value is an average on 100 simulations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

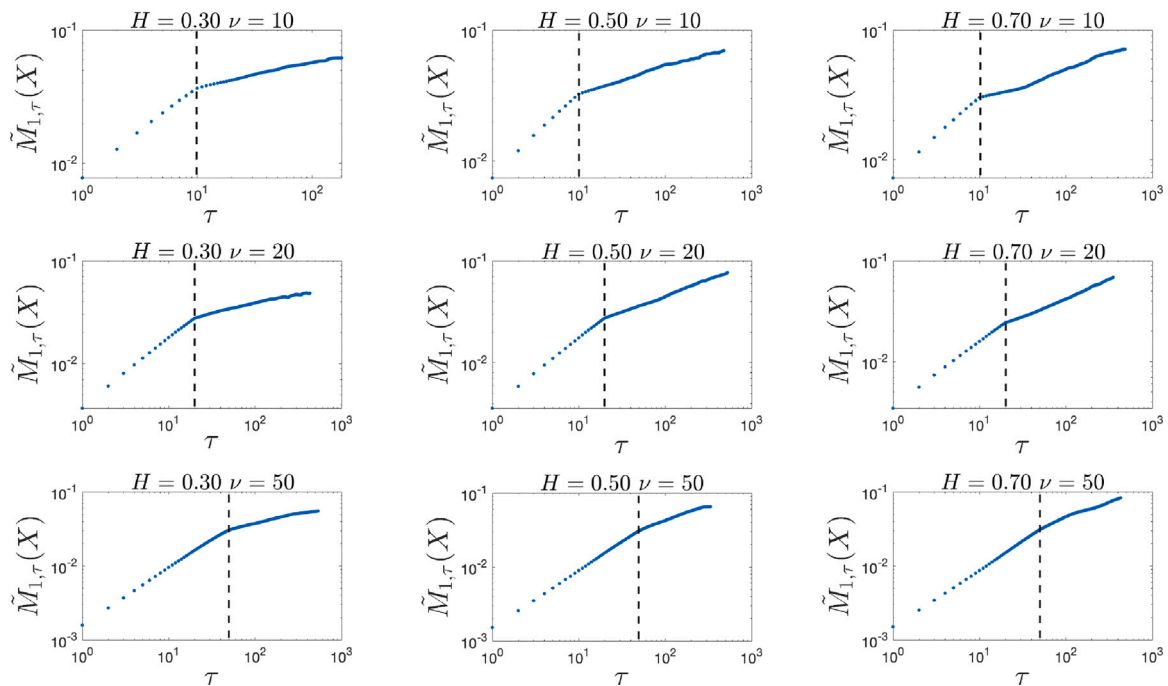


Fig. 5. Multiscaling analysis for $\hat{H}_t^{(\nu)}$, the estimated local regularity of MPRE, for various prescribed H exponent and length ν of LHE. $\hat{H}_t^{(\nu)}$ have two linear regions: the first, for $\tau \leq \nu$ where there are autocorrelations between data, and the second, for $\tau > \nu$ where autocorrelations are filtered out. Black dashed lines indicate when the temporal lag τ is equal to ν .

Finally, for the purpose of comparison with Table 1, the global Hurst exponents estimated from the GHE are reported in Table 2 for the same range of values of ν . Also for the GHE, the roughness increases as ν decreases and the standard errors are comparable with those of the HH method.

6. Conclusion

This work contributes to the current research on (eventually rough) stochastic volatility along the two main directions of modeling and estimation.

As to the first direction, the Multifractal Process with Random Exponent (MPRE) is examined as a robust candidate model of the

stock log-prices. The process is specified by its stochastic Hurst–Hölder parameter and behaves as a fractional Brownian motion (fBm) in a neighborhood of any point t_0 . Moving from empirical estimates, in past contributions (see, e.g. [19]), we outlined the characteristics that the stochastic Hurst exponent should display to ensure that the resulting MPRE replicates the same patterns shown by real financial time series. Roughly speaking, these are the stationarity, the mean reversion to the value 1/2, the nearly normal distribution and the presence of autocorrelation. These features naturally lead to choose the fractional Ornstein–Uhlenbeck process (fOU) to describe the dynamics of the Hurst parameter. Interestingly, if prices are modeled by an MPRE, this choice turns out to coincide with what is done by popular models of

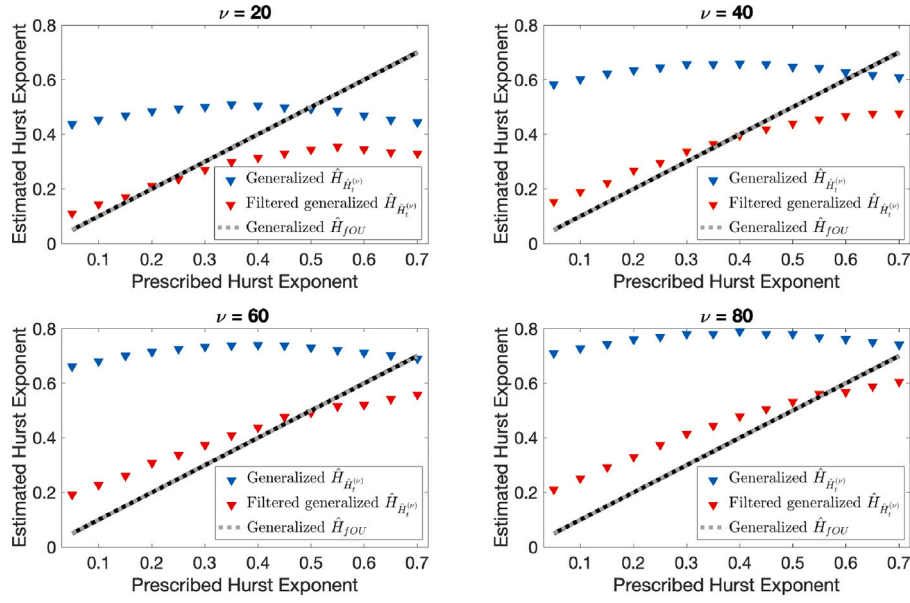


Fig. 6. Multiscaling analysis with temporal lags $\tau \in [1, \tau_{\max}]$ and different lengths ν . In all cases, $\hat{H}_{H_t}$ (gray dotted line) is superimposed to the bisector (black line), which indicates an almost perfect match between prescribed and estimated exponents. Conversely, both the filtered ($\tau \in [\nu + 1, \tau_{\max}]$, red triangles) and the unfiltered ($\tau \in [1, \tau_{\max}]$, blue triangles) estimates $\hat{H}_{H_t^{(0)}}$ display nonlinear patterns far from the bisector. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

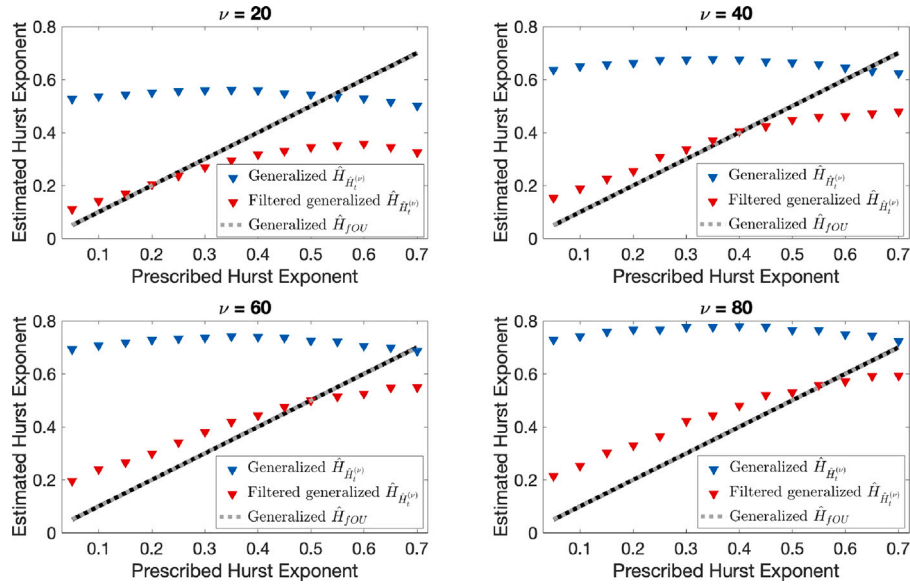


Fig. 7. Multiscaling analysis with temporal lags $\tau \in [1, \tau_{\max}]$ and different lengths ν for RNSGHE method. In all cases, $\hat{H}_{H_t}$ (gray dotted line) is superimposed to the bisector (black line), which indicates an almost perfect match between prescribed and estimated exponents. For the filtered ($\tau \in [\nu + 1, \tau_{\max}]$, red triangles) and the unfiltered ($\tau \in [1, \tau_{\max}]$, blue triangles) estimates $\hat{H}_{H_t^{(0)}}$, we obtain nonlinear patterns similar to the GHE method. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2

Global Hurst exponent from GHE $\hat{H}_{H_t^{(0)}}$ for seven stock indexes, compared with $\hat{H}_{H_t^{(0)}}$, ($\nu = 5, 10, 20$), obtained by applying estimator (30) to the MPRE surrogated with random parameters given by the sequences $\{(t, \hat{H}_t^{(0)})\}$ centered on 0.5. The values of $\tau_{\max}$ are estimated by using the ACSR method described in Section 3.2. Numbers in parenthesis represent the standard errors related to the estimated coefficients.

	$\tau_{\max}$	$\hat{H}_{H_t^{(0)}}$	$\hat{H}_{H_t^{(5)}}$	$\hat{H}_{H_t^{(10)}}$	$\hat{H}_{H_t^{(20)}}$
SPX	358	0.0915	0.085(0.002)	0.120(0.003)	0.151(0.005)
DJI	346	0.0910	0.088(0.002)	0.119(0.003)	0.153(0.004)
FTSE	331	0.1004	0.077(0.002)	0.105(0.002)	0.125(0.005)
HSI	321	0.0938	0.082(0.002)	0.120(0.003)	0.162(0.005)
N225	229	0.0859	0.072(0.002)	0.102(0.003)	0.129(0.004)
NSEI	327	0.0815	0.077(0.002)	0.101(0.003)	0.122(0.003)
SSMI	256	0.1457	0.085(0.002)	0.119(0.003)	0.151(0.004)

stochastic volatility such as the FSVM and the RFSVM. The reason resides in the inverse linear law which relates the log-volatility and the Hurst exponent of a fBm, whose stochastic signature is locally replicated by the MPRE. Thus, once a proper adjustment is made for its parameters, the fOU which models the log-volatility in (1) or in (2) can be used to model the stochastic Hurst-Hölder exponent to be plugged in the MPRE. Since an Hurst-Hölder exponent equal to $1/2$ indicates that the fractional process is a martingale, replacing the log-volatility by the Hurst parameter allows to evaluate when the examined time series departs from the martingale paradigm embodied by the Efficient Market Hypothesis, information that per se the log-volatility does not provide. Furthermore, since the Hurst-Hölder exponent is assumed to follow a predictable stochastic process, a further line of research could regard the forecast of the exponent to assess how far the price is from the equilibrium and the consequent probability of a correction to eliminate the inefficiency.

As to the second direction, we have estimated the overall Hurst parameter of the fOU by three different approaches: the Local Hurst Exponent (LHE), the Generalized Hurst Exponent (GHE) and the realized volatility. By using both surrogated and real financial time series, we provide experimental evidence that several factors can produce even large biases in the estimation of a (eventually rough) Hurst exponent. The biases are triggered by the size of the estimation window and by the choice of the parameters considered for the global estimator of the Hurst exponent. In particular, when the size of the window of the LHE estimator reduces to try to capture the pointwise parameter (i.e. the instantaneous volatility), the variance of the estimator can cause artificial roughness in the sequence. Using both LHE and GHE, the same pattern is observed for the realized volatility data. Of course, these findings do not rule out the possibility that the Hurst-Hölder process (respectively, the volatility process) may in fact be rough in any case; rather, they only provide a cautionary warning about the reliability of the results obtained with such analysis techniques.

It seems to us that further promising developments on this topic could concern the introduction of jumps in the MPRE, the analysis of the negative dependence found by [22] between prices' multiscaling and the Hurst exponent of the volatility process, and the estimation of the global Hurst exponent through a distribution-based method introduced by [60] and made robust with respect to dependent data by [61].

CRedit authorship contribution statement

Daniele Angelini: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Sergio Bianchi:** Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] Comte F, Renault E. Long memory in continuous-time stochastic volatility models. *Math Finance* 1998;8(4):291–323.
- [2] Gatheral J, Jaisson T, Rosenbaum M. Volatility is rough. *Quant Finance* 2018;18(6):933–49.
- [3] Livieri G, Mouti S, Pallavicini A, Rosenbaum M. Rough volatility: Evidence from option prices. *IJSE Trans* 2018;50(9):767–76.
- [4] Fukasawa M, Takabatake T, Westphal R. Is volatility rough?. 2019, 1905.04852, arXiv.org.
- [5] Takaishi T. Rough volatility of bitcoin. *Finance Res Lett* 2020;32:101379.
- [6] Bennedsen M, Lunde A, Pakkanen MS. Decoupling the short- and long-term behavior of stochastic volatility. *J Financ Econ* 2021;20(5):961–1006.
- [7] Bolko E, Christensen K, Pakkanen MS, Veliyeva B. A GMM approach to estimate the roughness of stochastic volatility. *J Econometrics* 2022. Available online 15 September 2022.
- [8] Fukasawa M. Volatility has to be rough. *Quant Finance* 2021;21(1):1–8.
- [9] Cont R, Das P. Rough volatility: Fact or artefact?. 2022, Available at SSRN: <https://ssrn.com/abstract=4065951>.
- [10] Rogers LCG. Lecture notes in finance options - 45 years since the publication of the black-scholes-merton model. In: Chapter things we think we know. *World Scientific*; 2023, p. 173–84.
- [11] Ayache A. Continuous Gaussian multifractional processes with random pointwise Hölder regularity. *J Theoret Probab* 2013;26:72–93.
- [12] Loboda D, Mies F, Steland A. Regularity of multifractional moving average processes with random hurst exponent. *Stochastic Process Appl* 2021;140:21–48.
- [13] Ayache A, Bouly F. Moving average multifractional processes with random exponent: Lower bounds for local oscillations. *Stochastic Process Appl* 2022;146:143–63.
- [14] Cajueiro DO, Tabak BM. The hurst exponent over time: testing the assertion that emerging markets are becoming more efficient. *Physica A* 2004;336(3–4):521–37.
- [15] Bianchi S, Pantanella A. Pointwise regularity exponents and well-behaved residuals in stock markets. *Int J Trade Econ Finance* 2011;2(1):52–60.
- [16] Peng Q, Zhao R. A general class of multifractional processes and stock price informativeness. *Chaos Solitons Fractals* 2018;115:248–67.
- [17] Garcin M. Forecasting with fractional Brownian motion: a financial perspective. *Quant Finance* 2022;22(8):1495–512.
- [18] Mattered R, Di Sciorio F, Trinidad-Segovia JE. A composite index for measuring stock market inefficiency. *Complexity* 2022;9838850, Special Issue: Complexity Arising in Financial Modelling and its Applications, 13 pages.
- [19] Bianchi S, Pantanella A, Pianese A. Modeling stock prices by multifractional Brownian motion: an improved estimation of the pointwise regularity. *Quant Finance* 2013;13(8):1317–30.
- [20] Ayache A, Taqqu MS. Multifractional processes with random exponent. *Publ Mat* 2005;49:459–86.
- [21] Cheridito P, Kawaguchi H, Maejima M. Fractional ornstein-uhlenbeck processes. *Electron J Probab* 2003;8:1–14.
- [22] Brandi G, Di Matteo T. Multiscaling and rough volatility: An empirical investigation. *Int Rev Financ Anal* 2022;84:102324.
- [23] Bianchi S, Pianese A. Multifractional properties of stock indices decomposed by filtering their pointwise Hölder regularity. *Int J Theor Appl Fin* 2008;11(6):567–95.
- [24] Fama EF. Efficient capital markets: A review of theory and empirical work. *J Finance* 1970;25(2):383–417.
- [25] Mandelbrot BB, Van Ness JW. Fractional Brownian motions, fractional noises and applications. *SIAM Rev* 1968;10(4):422–37.
- [26] Coeurjolly JF. Estimating the parameters of a fractional Brownian motion by discrete variations of its sample paths. *Stat Inference Stoch Process* 2001;4(2):199–227.
- [27] Rogers LCG. Arbitrage with fractional Brownian motion. *Math Finance* 1997;7(1):95–105.
- [28] Hu Y, Nualart D, Song J. Fractional martingales and characterization of the fractional Brownian motion. *Ann Probab* 2009;37(6):2404–30.
- [29] Peltier RS, Lévy Véhel J. Multifractional Brownian motion: Definition and preliminary results. *Rapport de Recherche INRIA* 1995;2645(1-39):1–39.
- [30] Ayache A, Lévy Véhel J. On the identification of the pointwise Hölder exponent of the generalized multifractional Brownian motion. *Stochastic Process Appl* 2004;111(1):119–56.
- [31] Benassi A, Jaffard S, Roux D. Gaussian processes and pseudodifferential elliptic operators. *Revista Math Iberoamericana* 1997;13:19–89.
- [32] Ayache A, Lévy Véhel J. The generalized multifractional Brownian motion. *Stat Inference Stoch Process* 2000;3(1–2):7–18.
- [33] Houdré C, Villa J. An example of infinite dimensional quasi-helix. *Contemp Math* 2003;336:195–201.
- [34] Cheridito P. Mixed fractional Brownian motion. *Bernoulli* 2001;7(6):913–34.
- [35] Anh VV, Angulo JM, Ruiz-Medina MD. Possible long-range dependence in fractional random fields. *J Statist Plan Inference* 1999;80:95–110.
- [36] Li M. Multi-fractional generalized Cauchy process and its application to teletraffic. *Physica A* 2020;550:123982.

- [37] Lim SC, Eab CH. Some fractional and multifractional Gaussian processes: A brief introduction. *Int J Modern Phys Conf Series* 2015;36:1560001.
- [38] Ayache A, Esser C, Hamonier J. A new multifractional process with random exponent. *Risk Decis Anal* 2018;7(1–2):5–29.
- [39] Corlay S, Lebovitis J, Lévy Véhel J. Multifractional stochastic volatility models. *Math Finance* 2014;24:364–402.
- [40] Bennesen M, Lunde A, Pakkanen MS. Decoupling the short- and long-term behavior of stochastic volatility. *J Financ Econom* 2022;20:961–1006.
- [41] Bianchi S, Pantanella A, Pianese A. Efficient markets and behavioral finance: a comprehensive multifractional model. *Adv Complex Syst* 2015;18(1–2):1550001:1–29.
- [42] Iatas J, Lang G. Variations quadratiques et estimation de l'exposant de hölder local d'un processus gaussien. *Ann Inst Henri Poincaré* 1997;33(4):407–36.
- [43] Kent JT, Wood ATA. Estimating the fractal dimension of a locally selfsimilar Gaussian process using increments. *J R Stat Soc Series B* 1997;59(3):679–700.
- [44] Benassi A, Bertrand P, Cohen S, Iatas J. Identification of the hurst index of a step fractional Brownian motion. *Stat Inference Stoch Process* 2000;3(1–2):101–11.
- [45] Pianese A, Bianchi S, Palazzo AM. Fast and unbiased estimator of the time-dependent hurst exponent. *Chaos* 2018;28(31102):1–6.
- [46] Bianchi S. Pathwise identification of the memory function of the multifractional Brownian motion with application to finance. *Int J Theor Appl Finance* 2005;8(2):255–81.
- [47] Taqqu MS, Teverovsky V, Willinger W. Estimators for long-range dependence: an empirical study. *Fractals* 1995;3(4):785–98.
- [48] Beran J. Statistics for long memory processes. Chapman and Hall, London; 1994.
- [49] Higuchi T. Approach to an irregular time series on the basis of the fractal theory. *Physica D* 1988;31:277–83.
- [50] Barabási AL, Vicsek T. Multifractality of self-affine fractals. *Phys Rev A* 1991;44(4):2730.
- [51] Di Matteo T. Multi-scaling in finance. *Quant Finance* 2007;7(1):21–36.
- [52] Buonocore RJ, Aste T, Di Matteo T. Measuring multiscaling in financial time-series. *Chaos Solitons Fractals* 2016;88:38–47.
- [53] Buonocore RJ, Aste T, Di Matteo T. Asymptotic scaling properties and estimation of the generalized hurst exponents in financial data. *Phys Rev E* 2017;95:042311.
- [54] Yue P, Xu H-C, Chen W, Xiong X, Zhou W-X. Linear and nonlinear correlations in the order aggressiveness of Chinese stocks. *Fractals* 2017;25(05):1750041.
- [55] Zhi-Qiang J, Xie W-J, Zhou W-X, Sornette D. Multifractal analysis of financial markets: A review. *Rep Progr Phys* 2019;82:125901.
- [56] Brandi G, Di Matteo T. On the statistics of scaling exponents and the multiscaling value at risk. *Eur J Finance* 2022;28:1361–82.
- [57] Belfadli R, Es-Sebaïy K, Ouknine Y. Parameter estimation for fractional Ornstein–Uhlenbeck processes: non-ergodic case. *Front Sci Eng* 2011;1:1–16.
- [58] Hu Y, Nualart D. Parameter estimation for fractional Ornstein–Uhlenbeck processes. *Statist Probab Lett* 2010;80(11):1030–8.
- [59] Garcin M. Fractal analysis of the multifractality of foreign exchange rates. *Math Methods Econ* 2019;13/14(1):49–74.
- [60] Bianchi S. A new distribution-based test of self-similarity. *Fractals* 2004;12(03):331–46.
- [61] Chicheportiche R, Bouchaud J-P. Weighted Kolmogorov-Smirnov test: Accounting for the tails. *Phys Rev E* 2012;86:041115.

